Mechanochemical Processing of Organic Matter During Asteroid Gardening

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Abstract

Carbonaceous chondrites contain up to _˜4 wt% organic material, much of which likely derives from precursors formed in the protosolar nebula or parent molecular cloud (1). Although subsequent hydrothermal and radiative evolution modified this inventory, shock-related processes remain poorly understood. Studies of carbon-rich "rubble-pile" asteroids indicate that they form and evolve through successive impact and reaccumulation events. Moreover, analyses of returned samples show that these bodies efficiently absorb and dissipate shock energy, localising deformation and allowing them to persist for millions of years (2). Hexamethylenetetramine (HMT) and derivatives form in UV-irradiated laboratory analogues of interstellar ices (3), and HMT has been identified in extracts of the Murchison meteorite (4). Under hydrothermal conditions, HMT breaks down to ammonia and formaldehyde (5), key intermediates in soluble organic synthesis in extraterrestrial materials. Here we investigate the solid-state mechanochemistry of HMT with sodium-rich montmorillonite (MMT) to assess how shock/impact- driven processes influence organic evolution. HMT undergoes complex chemistry under mechanical stress, especially when the clay structure is disrupted, producing compounds including HMT-related species (e.g., HMT-CH, HMT-OH), pyrazine, and triazinane/triazine derivatives. Oxygen-bearing products form even without liquid water. In contrast, when the clay structure is preserved, mechanical energy induces intercalation of HMT-like species (HMT, HMT CH, and a triazinane derivative) into the clay interlayer space. The "cyclic mechanical input" might be applied as an analogue for micrometeorite and

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IDP impacts on asteroid surfaces. Our findings demonstrate that shock and impact events can influence the transformation and preservation of extraterrestrial organic matter, providing new insight into how impact and shock processes may have shaped the organic inventory of small bodies and influenced the exogenous delivery of prebiotic molecules to planetary environments. (1) Glavin et al., Primitive Meteorites and Asteroids. 2018. pp. 205–271; (2) Robin, C.Q. et al., Nat Commun, 2024, 15, 6203; (3) Vinogradoff et al., A&A, 2011, 530, A128; (4) Oba et al., Nat Commun, 2020, 11, 6243; (5) Vinogradoff et al., Geochim. Cosmochim. Acta, 2020, 269, 150–166